

week_12\week12\src\main\java\week12\Board.java

```
1 package week12;
2
3 /**
4  * An abstract class representing the game board for tic-tac-toe.
5  */
6 public abstract class Board {
7
8     /**
9      * Prints the current state of the board.
10     */
11     public abstract void print();
12
13     /**
14      * Returns the total number of cells on the board.
15      * @return
16      */
17     public abstract int getTotalCells();
18
19     /**
20      * Checks if the board is full.
21      * @return
22      */
23     public abstract boolean isBoardFull();
24
25     /**
26      * Checks if a cell is empty.
27      * @param pos the position to check
28      * @return
29      */
30     public abstract boolean isEmpty(Position pos);
31
32     /**
33      * Returns the position of a cell based on its box number.
34      * @param boxNumber the box number
35      * @return
36      */
37     public abstract Position getCellPosition(int boxNumber);
38
39     /**
40      * Places a marker on the board at the specified position.
41      * @param pos the position to place the marker
42      * @param marker the marker to place
43      */
44     public abstract void placeMarker(Position pos, char marker);
45
46     /**
47      * Checks for a winner on the board.
48      * @return the marker of the winning player, or '\0' if no winner
49      */
50     public abstract char checkWinner();
51 }
```

```
52     /**
53      * Checks if there is a line of markers starting from a given position.
54      * @param startPos the starting position
55      * @param steps the steps to move in each direction
56      * @return true if there is a line of markers, false otherwise
57      */
58     protected abstract boolean checkLine(Position startPos, int... steps);
59
60     /**
61      * Loads the board state from a string representation.
62      * For example "000201000" represents a 3x3 board with the following layout:
63      * 0 0 0
64      * 2 0 1
65      * 0 0 0
66      * @param line the string representation of the board
67      */
68     protected abstract void loadString(String line);
69
70     /**
71      * Returns a string representation of the board for network transmission.
72      * For example, if a 3 x 3 board has the following layout:
73      * 0 0 0
74      * 2 0 1
75      * 0 0 0
76      * The string representation would be "000201000".
77      * @return the string representation of the board
78      */
79     protected abstract String networkString();
80 }
```